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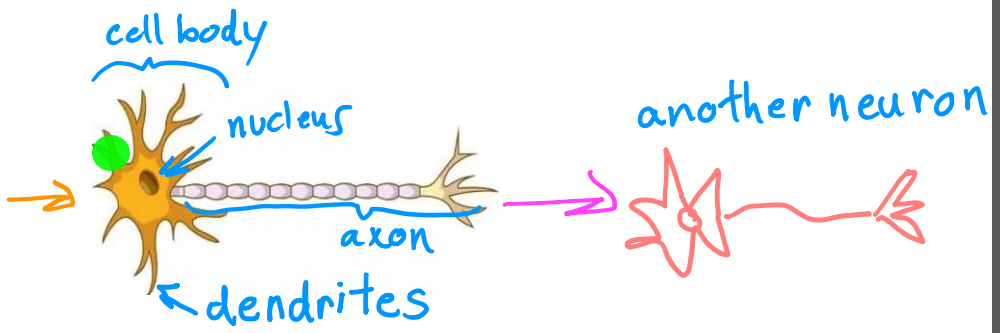
Neural Networks

Biological neuron

inputs outputs



Simplified mathematical model of a neuron

inputs outputs

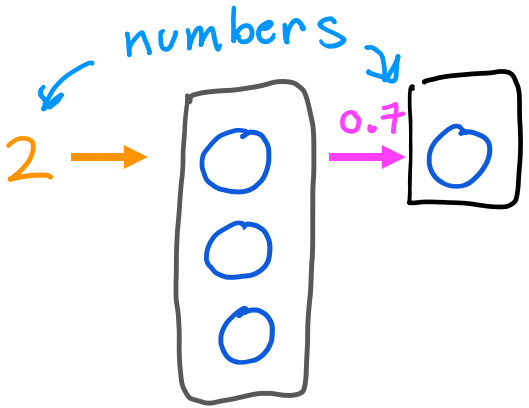
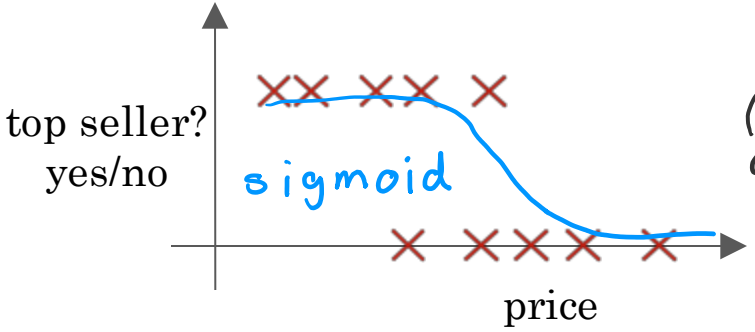


image source: https://biologydictionary.net/sensory-neuron/

Andrew Ng

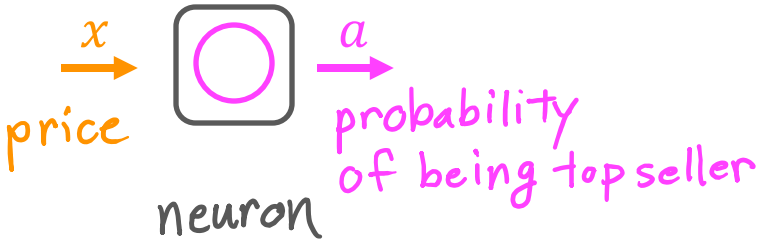
Demand Prediction



$x = price$ input

$a = f(x) = \dfrac{1}{1 + e^{-(wx+b)}}$ output

activation



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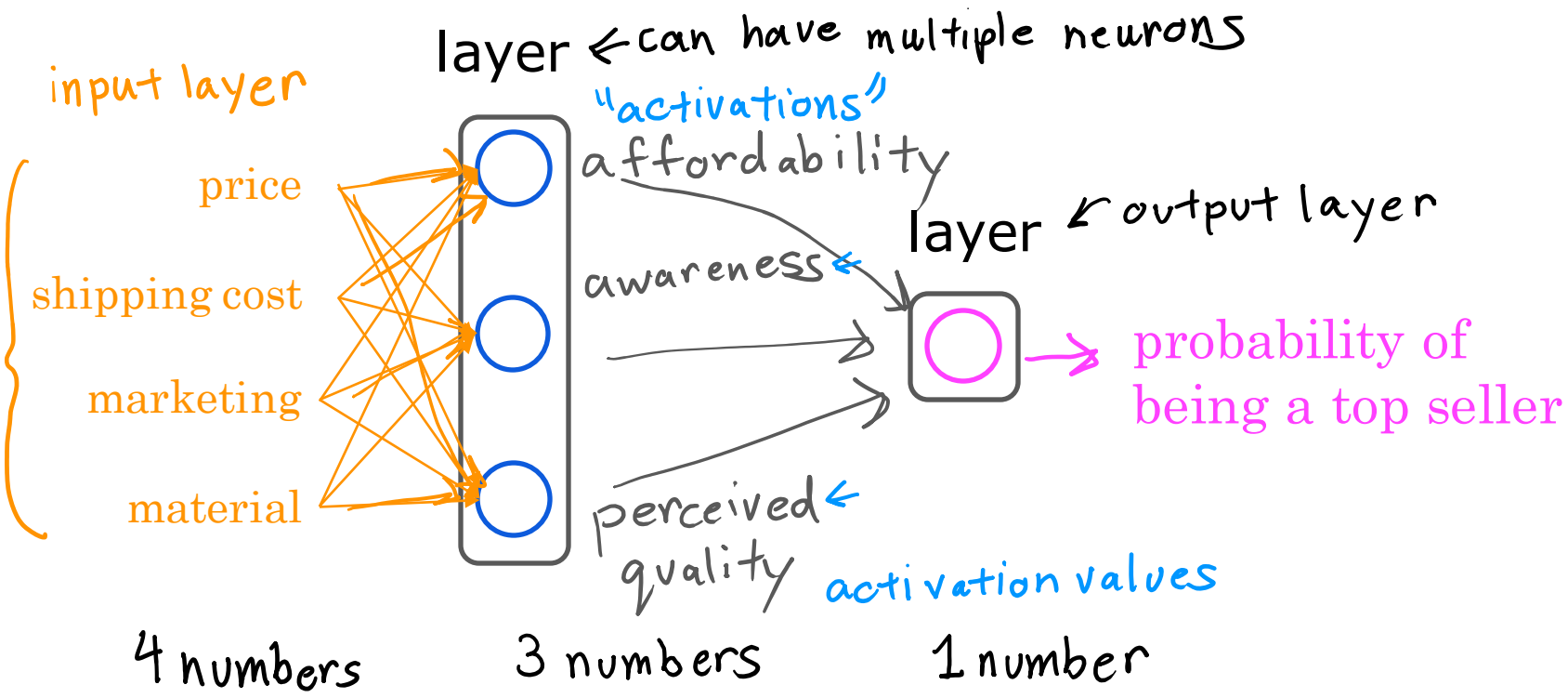
Demand Prediction



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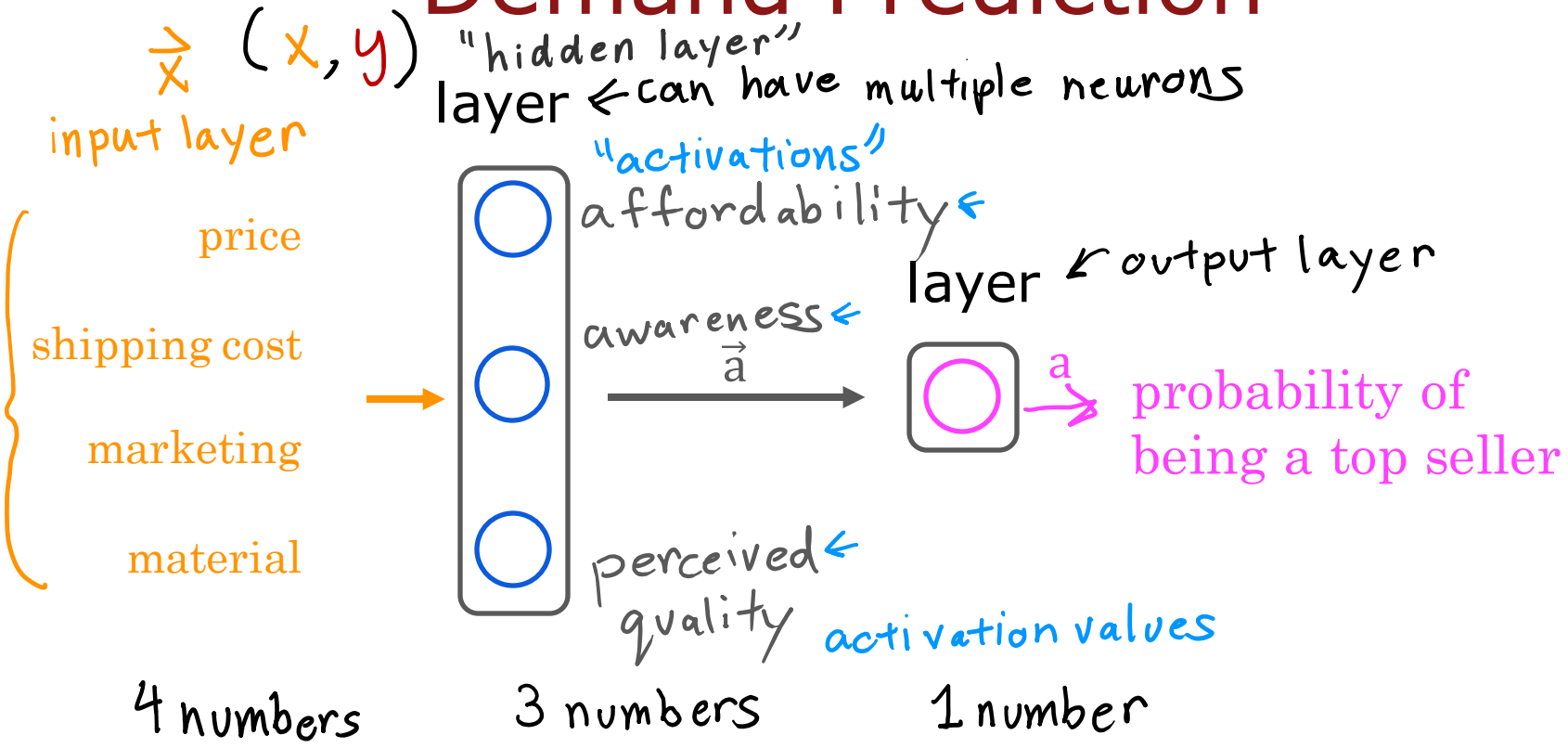
Demand Prediction



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Demand Prediction

feature engineering

x_1, x_2

$\vec{x}$ (x, y)

input layer

price

shipping cost

marketing

material

"hidden layer"

layer ← can have multiple neurons

"activations"

affordability ←

awareness ←

perceived quality ←

$\vec{a}$

layer ← output layer

a → probability of being a top seller

activation values

4 numbers

3 numbers

1 number

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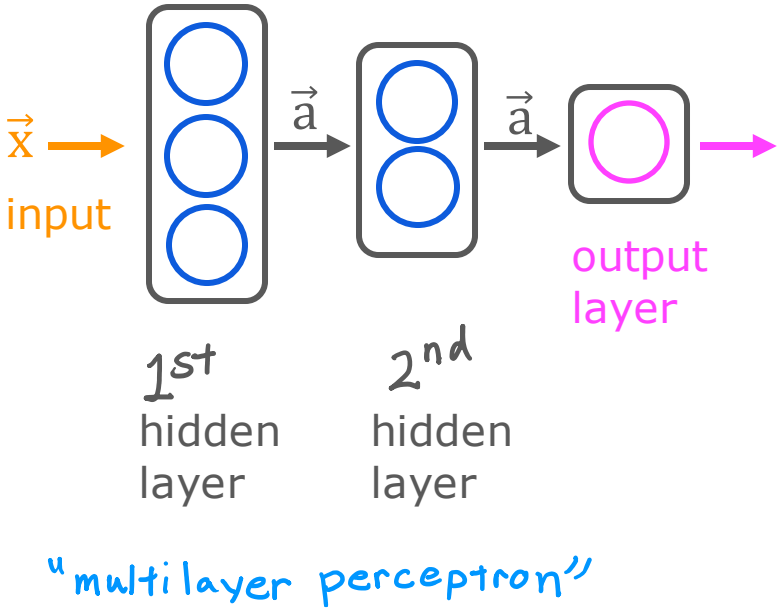
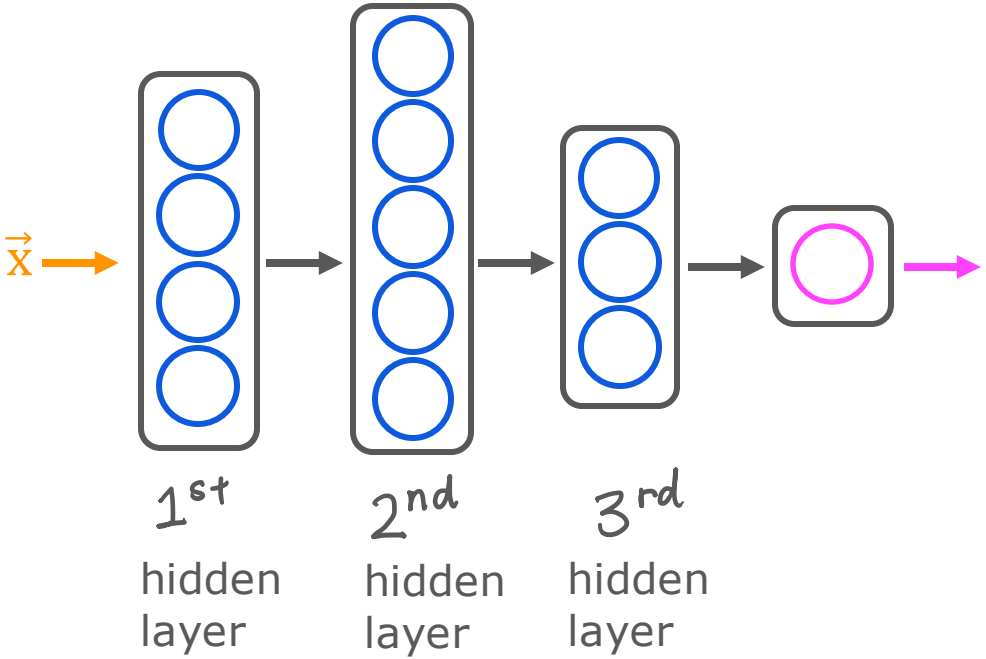
Multiple hidden layers





neural network architecture

Andrew Ng

Face recognition

1000 pixels

1000 pixels

1000 columns

1000 rows

197	185	203	...	...
...	57	64	92	...
⋮				
		...	187	214

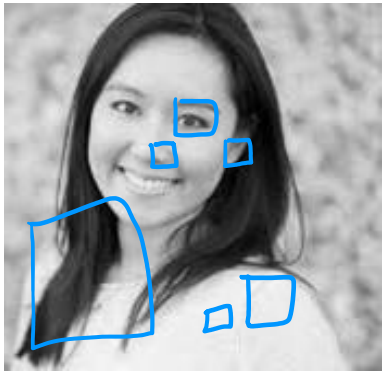
$\vec{x} =$

197
185
203
⋮
57
64
92
⋮
187
214

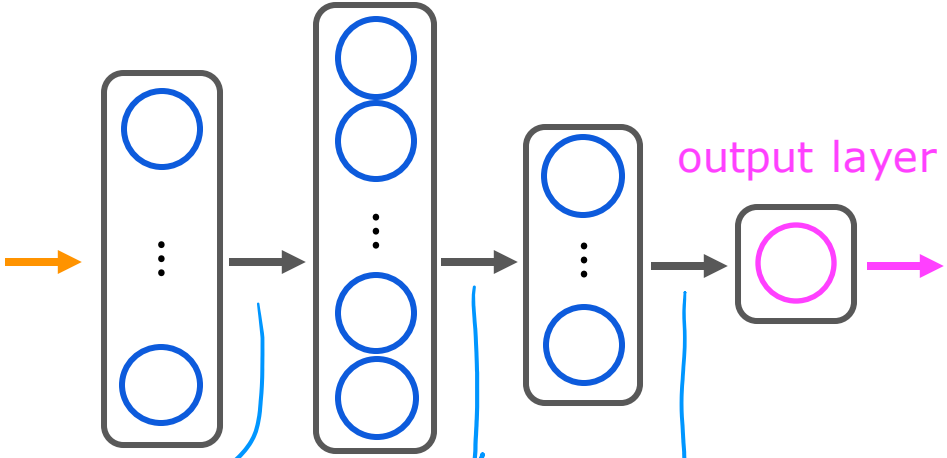
1 million

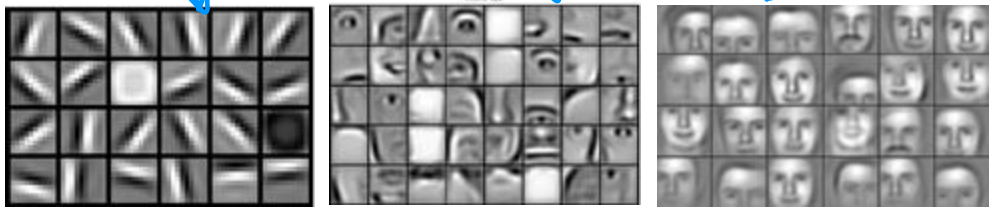
Andrew Ng

Face recognition



$\vec{x}$
input

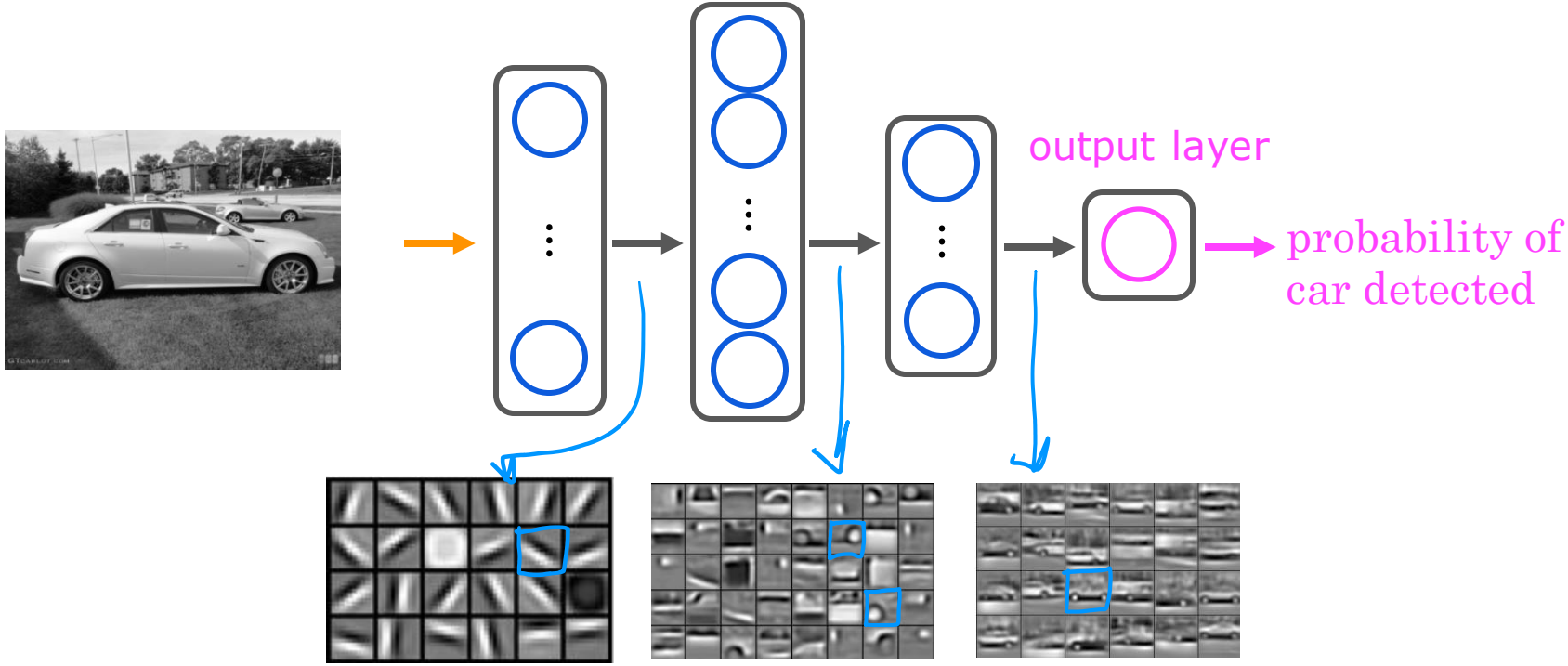




source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

Andrew Ng

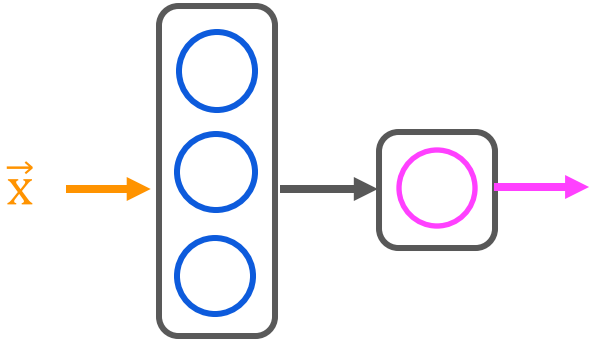
Car classification



source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

Andrew Ng

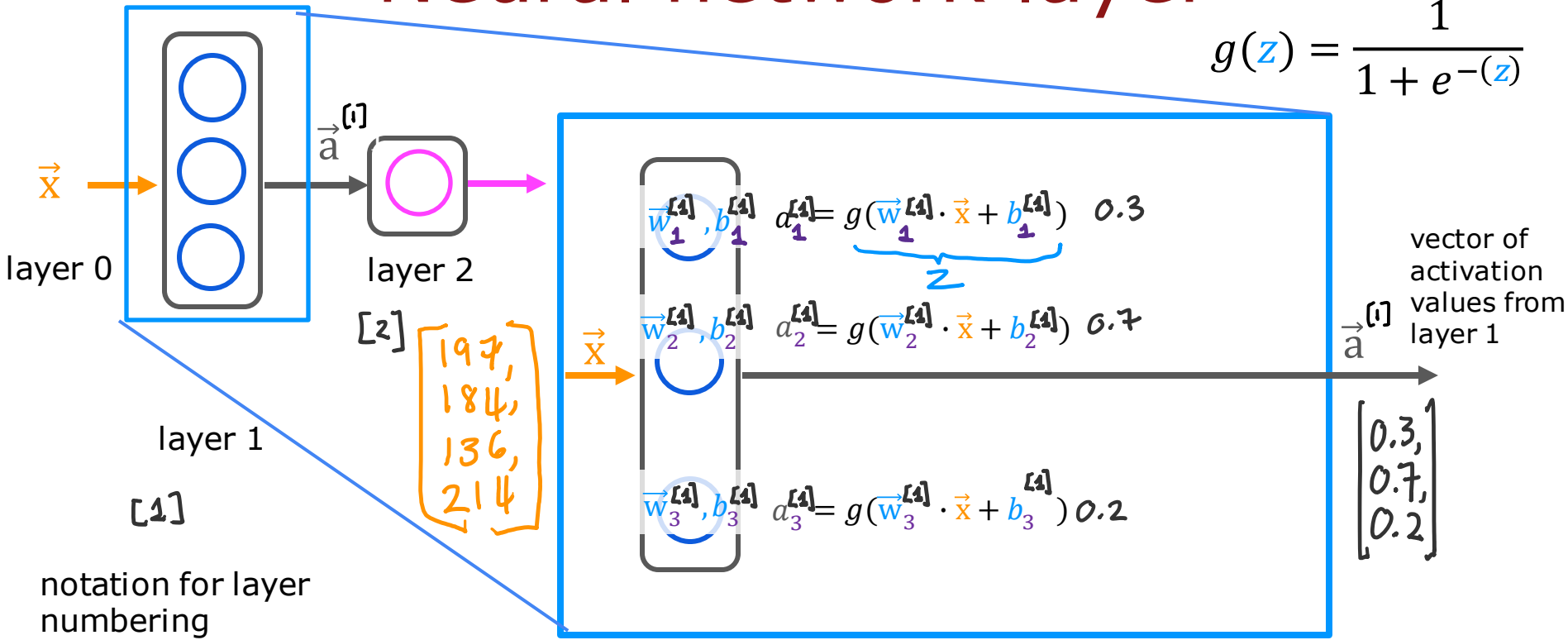
Neural network layer



Andrew Ng

Neural network layer

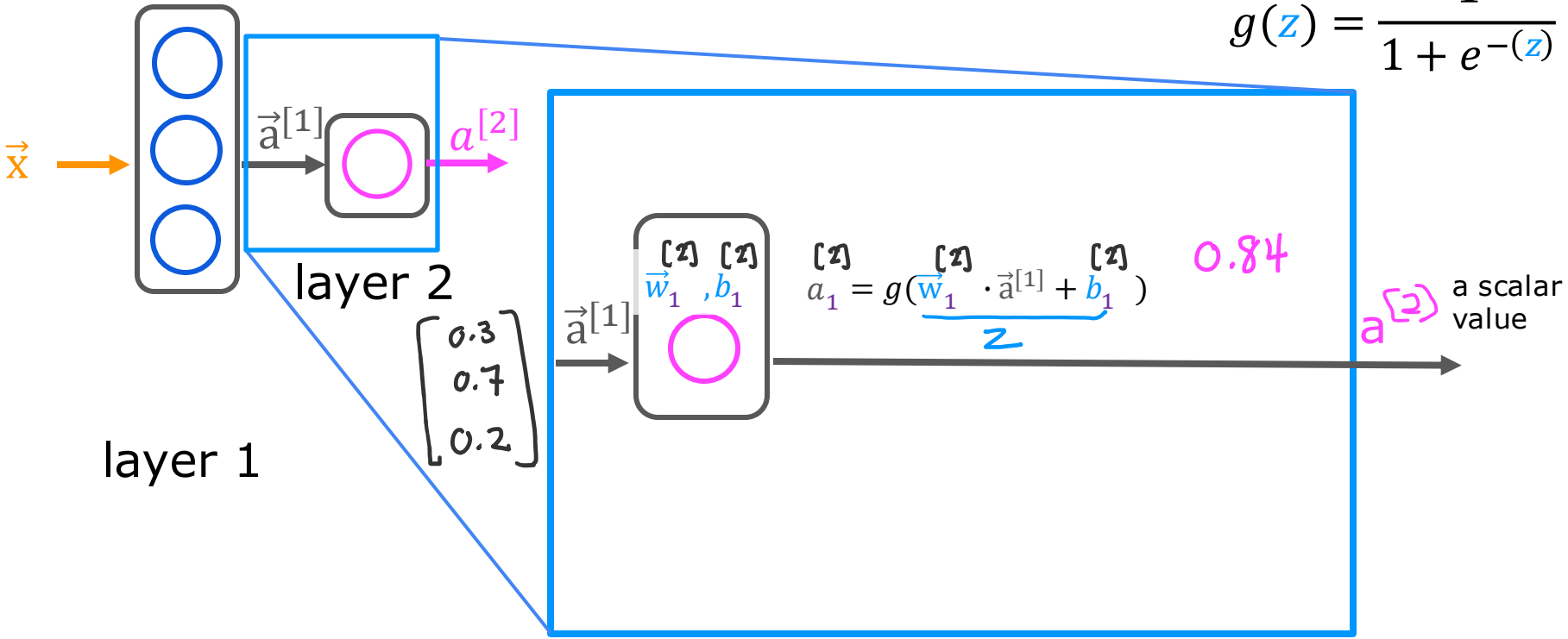
$$g(z) = \frac{1}{1 + e^{-(z)}}$$



Andrew Ng

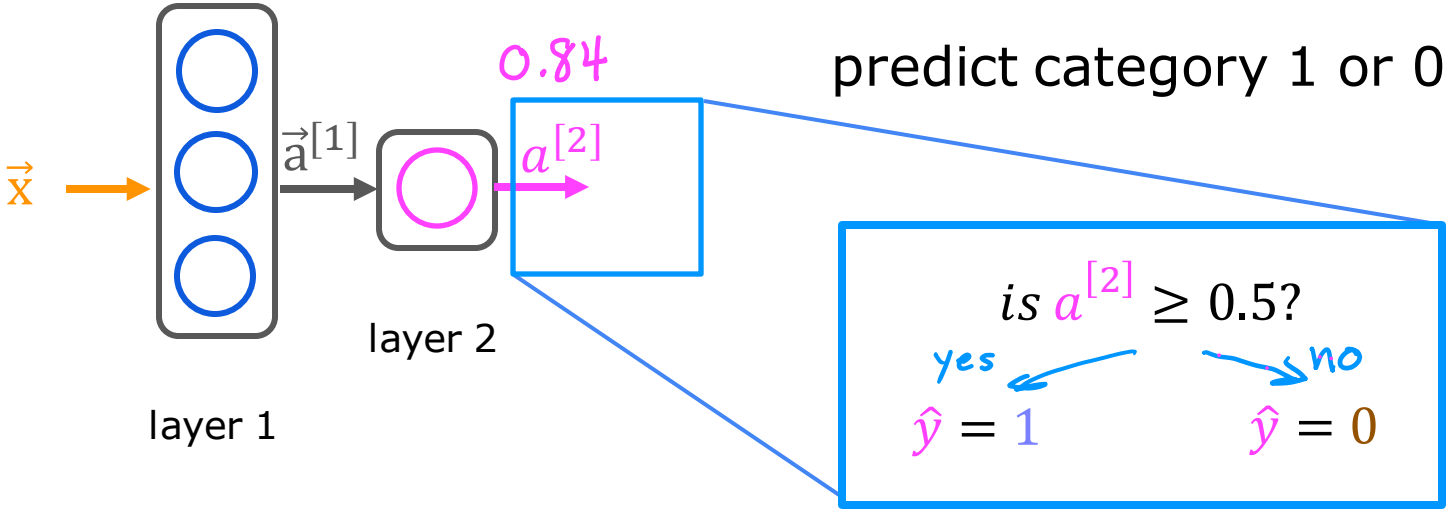
Neural network layer

$$g(z) = \frac{1}{1 + e^{-(z)}}$$



Andrew Ng

Neural network layer



predict category 1 or 0 (yes/no)

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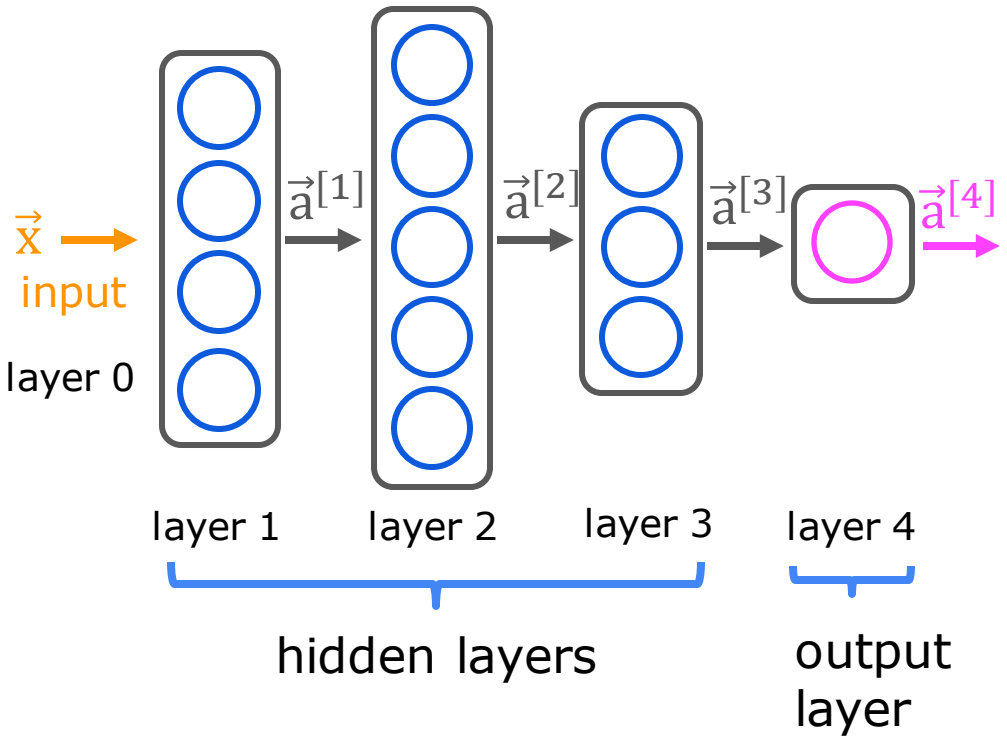
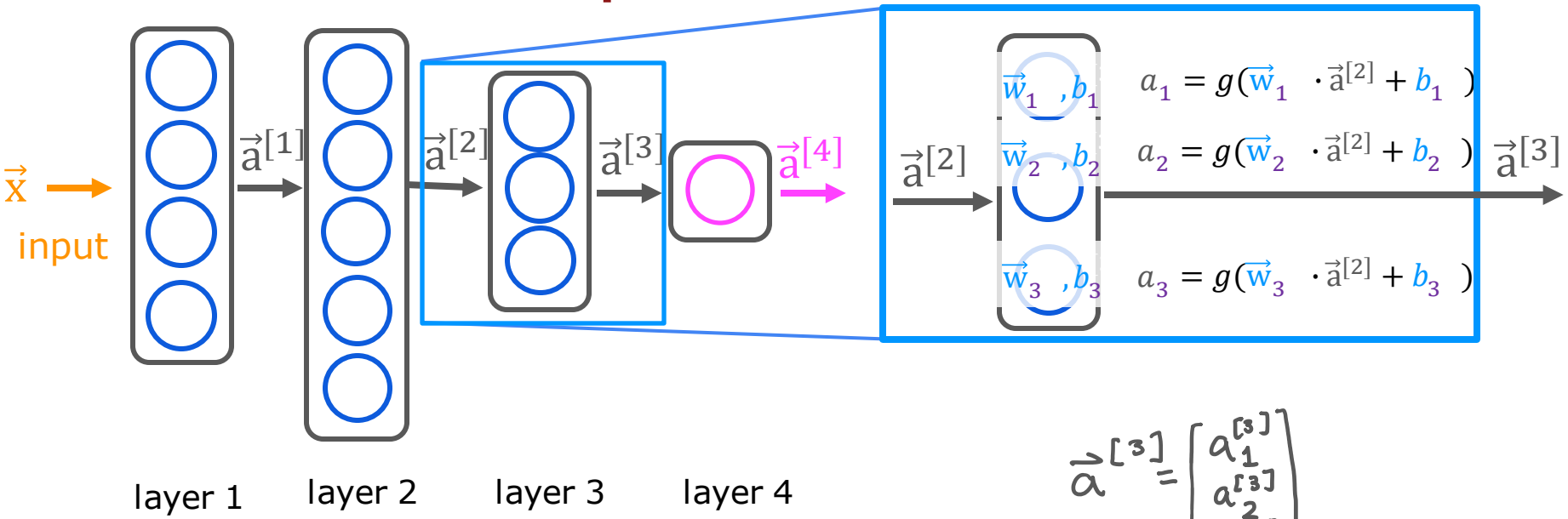
More complex neural network



Andrew Ng

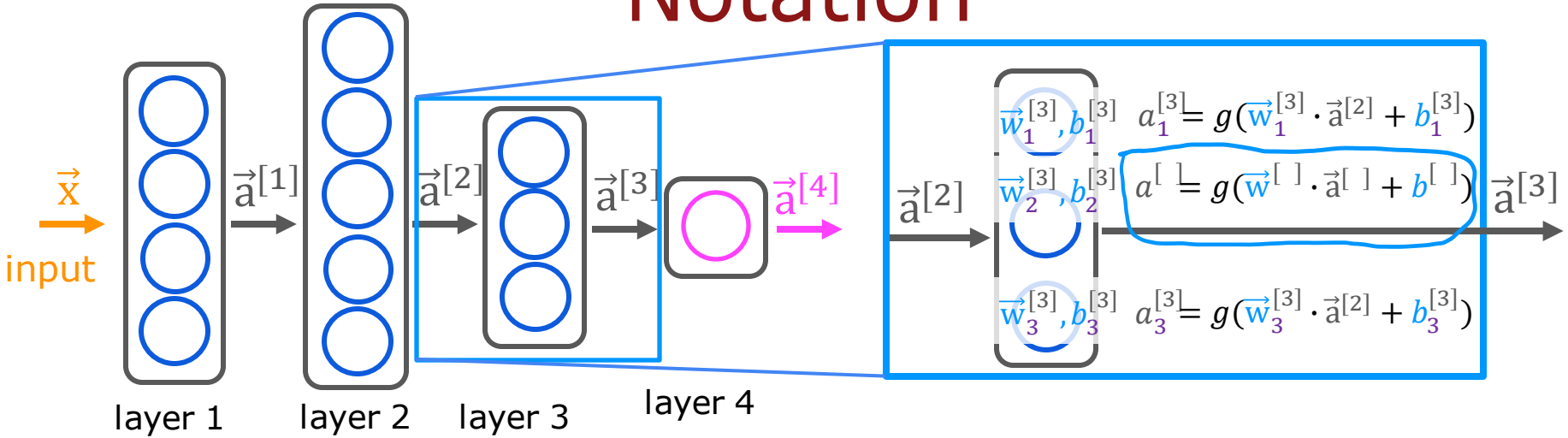
More complex neural network



$$\vec{a}^{[3]} = \begin{bmatrix} a_1^{[3]} \\ a_2^{[3]} \\ a_3^{[3]} \end{bmatrix}$$

Andrew Ng

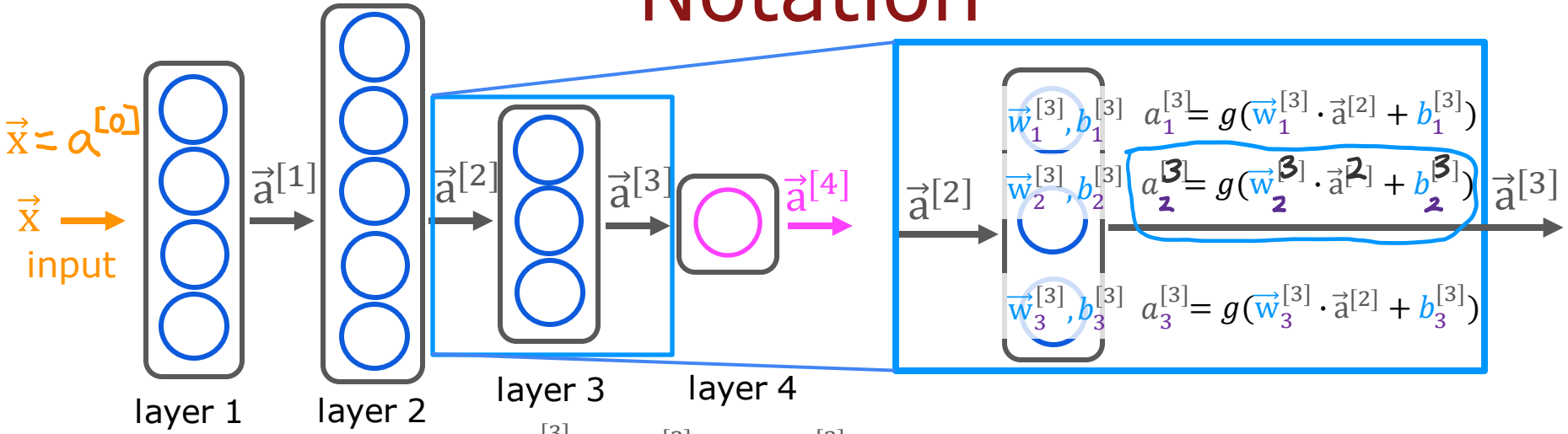
Notation



Question:
Can you fill in the superscripts and subscripts for the second neuron?

Andrew Ng

Notation



$$a_2^{[3]} = g(\vec{w}_2^{[3]} \cdot \vec{a}^{[2]} + b_2^{[3]})$$

Activation value of layer l, unit(neuron) j

output of layer $l-1$ (previous layer)

$$\boxed{a_j^{[l]} = g(\vec{w}_j^{[l]} \cdot \vec{a}^{[l-1]} + b_j^{[l]})}$$

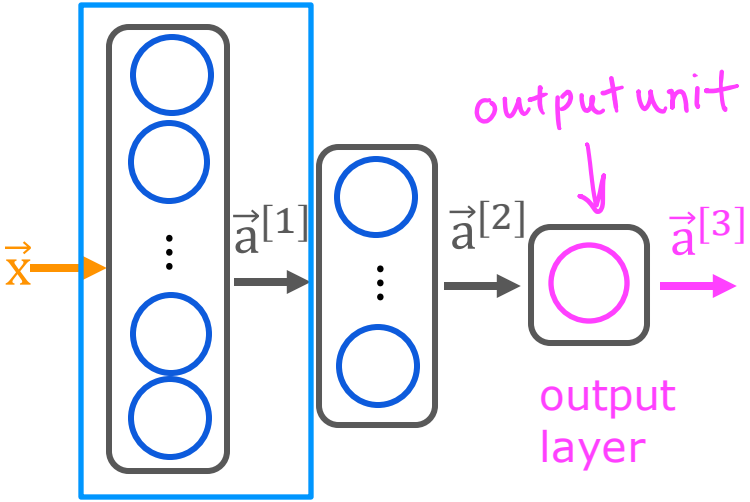
sigmoid

"activation function"

Parameters w & b of layer l, unit j

Andrew Ng

Handwritten digit recognition



probability of being a handwritten '1'

25 units — layer 1

15 units — layer 2

1 unit — layer 3

hidden units (neurons)

Digit images

label 0 1

255	255	255	255	255	255	255	255
255	255	255	0	255	255	255	255
255	255	0	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	0	0	0	255	255	255
255	255	255	255	255	255	255	255

8

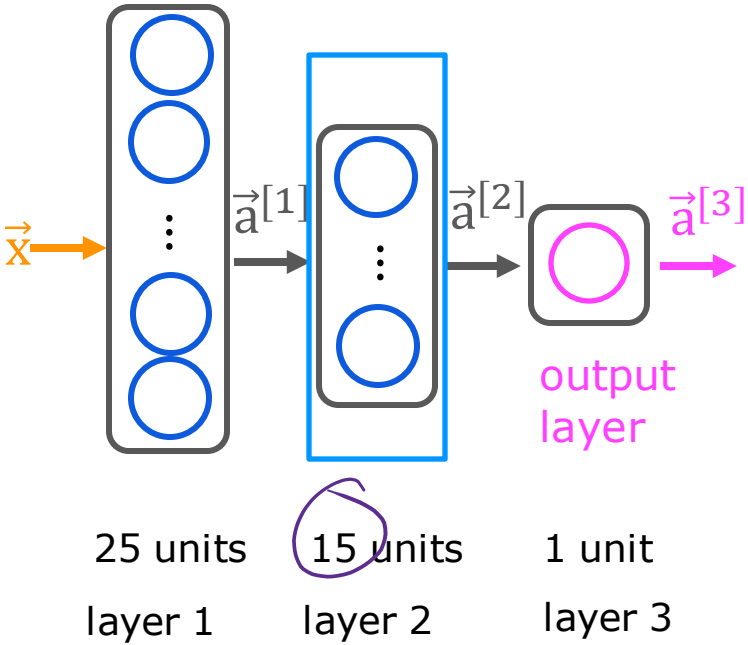
8

$a^{[0]}$

$$\vec{a}^{[1]} = \begin{bmatrix} g(\vec{w}_1^{[1]} \cdot \vec{x} + b_1^{[1]}) \\ \vdots \\ g(\vec{w}_{25}^{[1]} \cdot \vec{x} + b_{25}^{[1]}) \end{bmatrix}$$

Andrew Ng

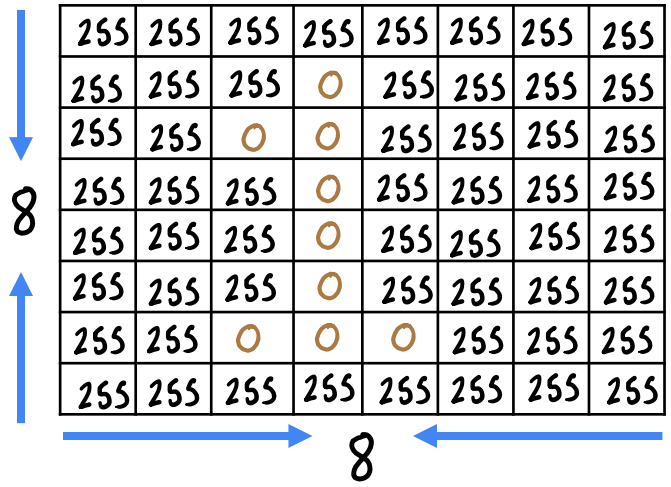
Handwritten digit recognition



probability of being a handwritten '1'

Digit images

label 0 1



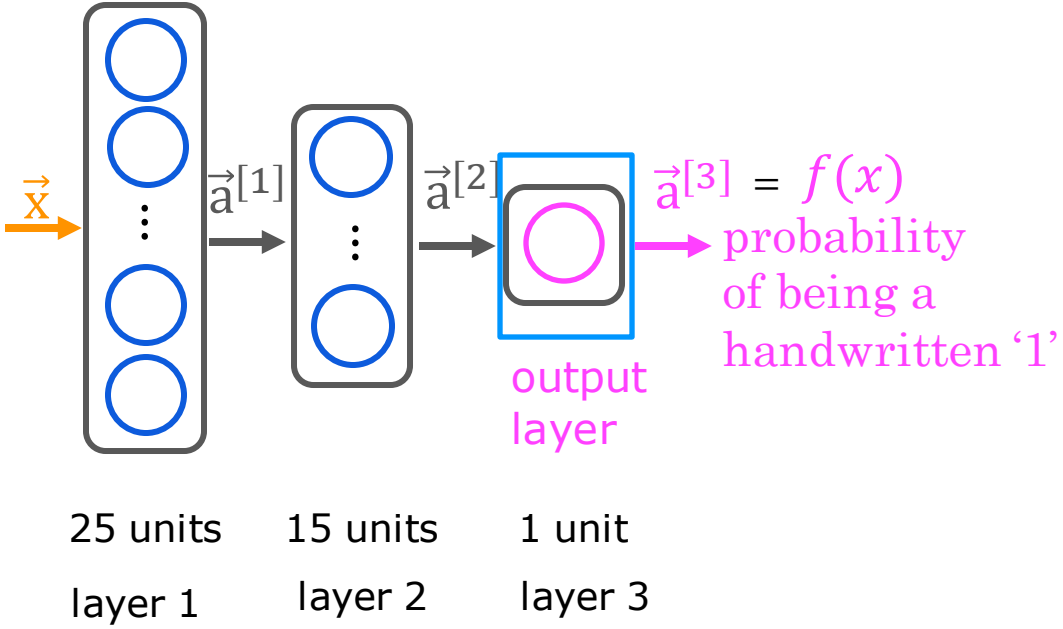
255	255	255	255	255	255	255	255
255	255	255	0	255	255	255	255
255	255	0	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	255	0	255	255	255	255
255	255	0	0	0	255	255	255
255	255	255	255	255	255	255	255

$$\vec{a}^{[2]} = \begin{bmatrix} g(\vec{w}_1^{[2]} \cdot \vec{a}^{[1]} + b_1^{[2]}) \\ \vdots \\ g(\vec{w}_{15}^{[2]} \cdot \vec{a}^{[1]} + b_{15}^{[2]}) \end{bmatrix}$$

Andrew Ng

Handwritten digit recognition

forward propagation



$$\vec{a}^{[3]} = \left[g\left(\vec{w}_1^{[3]} \cdot \vec{a}^{[2]} + b_1^{[3]} \right) \right]$$

$$is\ a_1^{[3]} \geq 0.5?$$

yes → $\hat{y} = 1$ image is digit 1

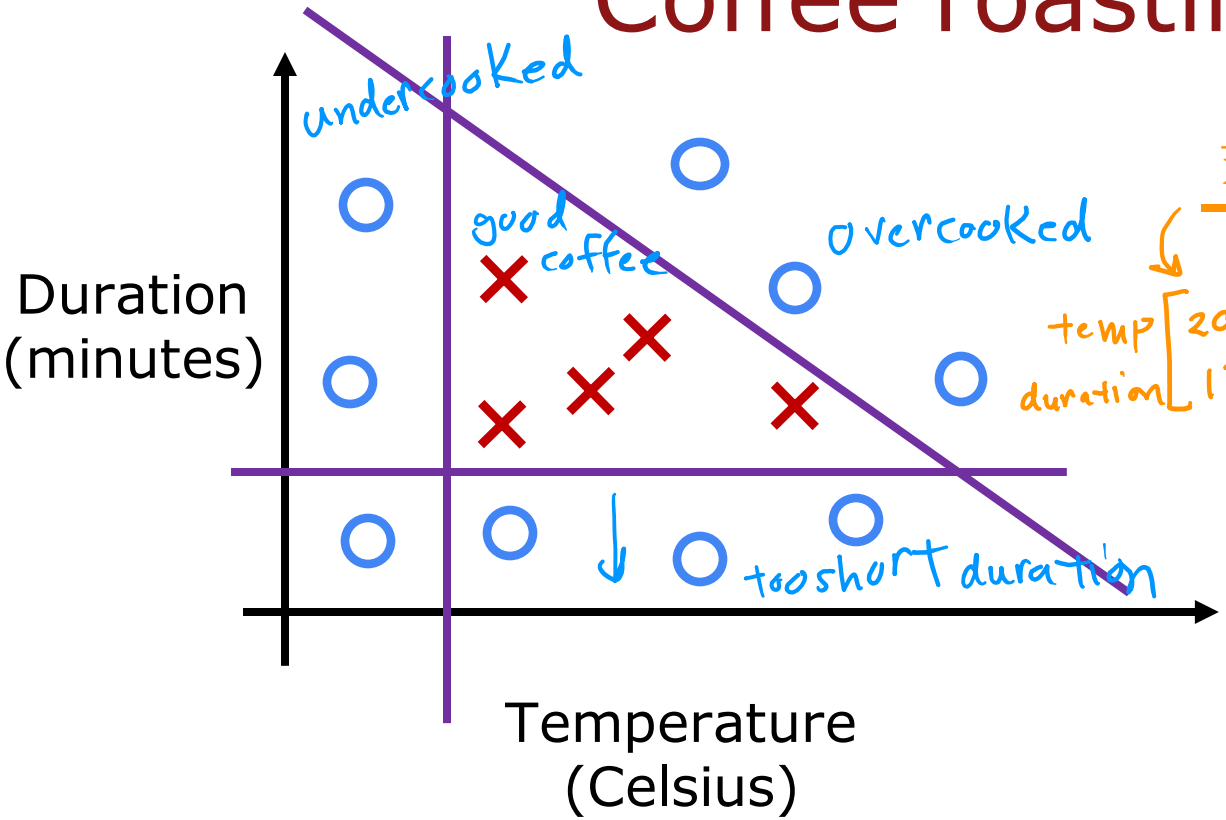
no → $\hat{y} = 0$ image isn't digit 1

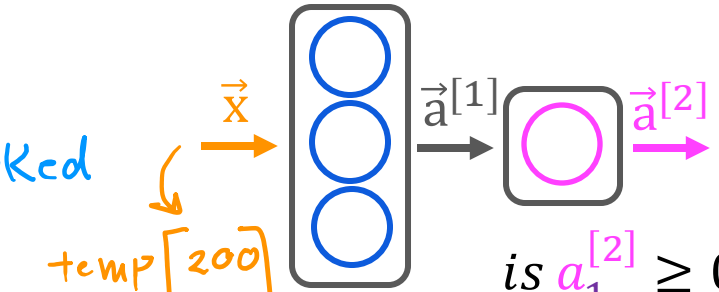
Andrew Ng

TensorFlow implementation

Inference in Code

Coffee roasting





$is\ a_1^{[2]} \geq 0.5?$

yes: $\hat{y} = 1$

no: $\hat{y} = 0$

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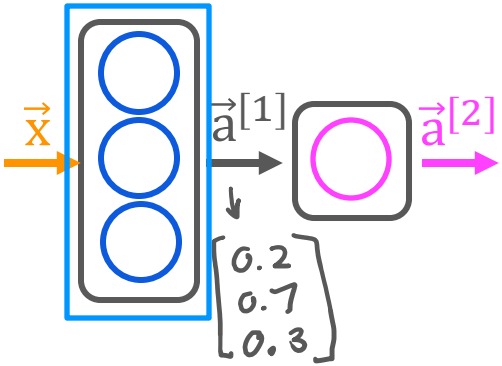
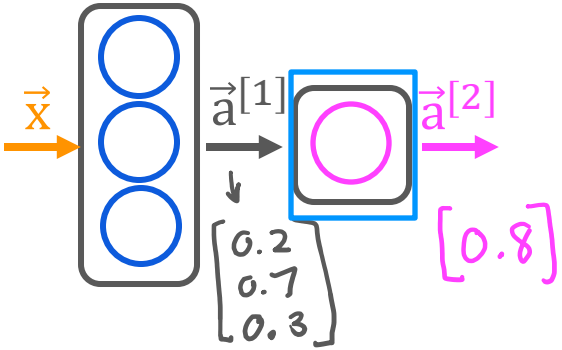



```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

Andrew Ng

Build the model using TensorFlow

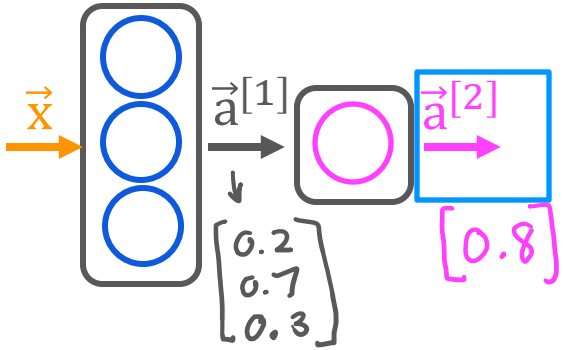



```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

Andrew Ng

Build the model using TensorFlow



$is\ a_1^{[2]} \geq 0.5?$

yes → $\hat{y} = 1$ ✗

no → $\hat{y} = 0$ ○

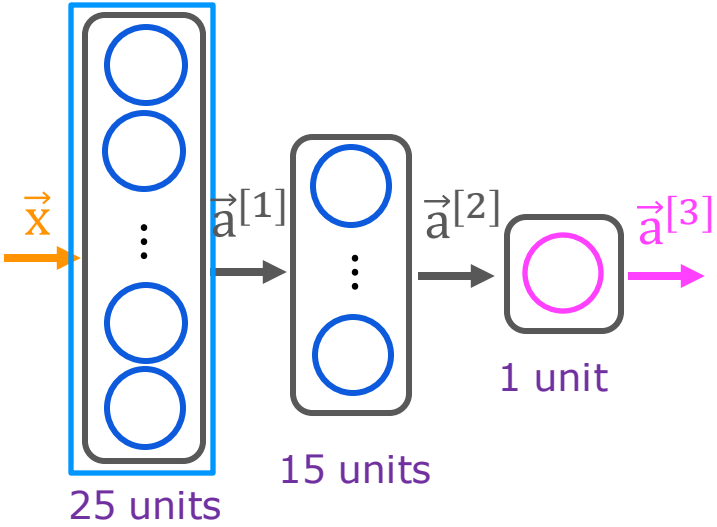
```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

```
if a2 >= 0.5:
  yhat = 1
else:
  yhat = 0
```

Andrew Ng

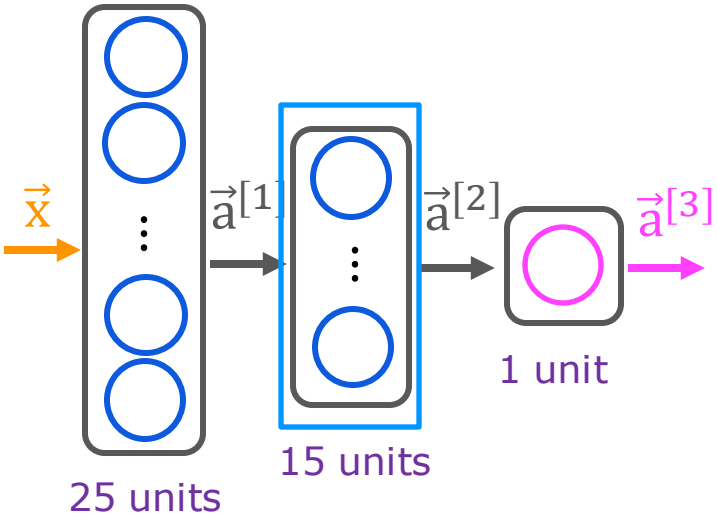
Model for digit classification




```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

Andrew Ng

Model for digit classification

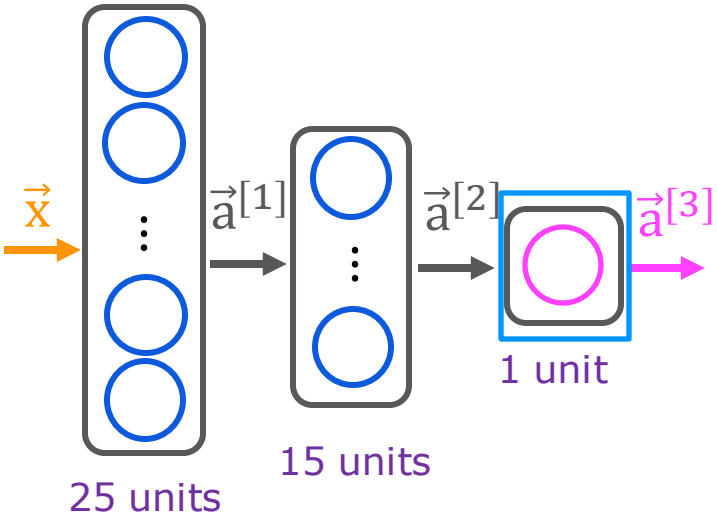



```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)
```

Andrew Ng

Model for digit classification



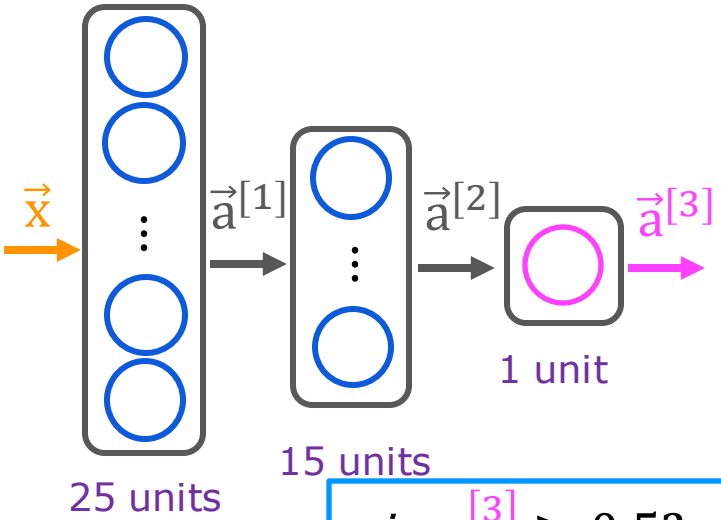

```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)

layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)

layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```

Andrew Ng

Model for digit classification




```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)

layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)

layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```

$is\ a_1^{[3]} \geq 0.5?$

$\hat{y} = 1$ ✗ $\hat{y} = 0$ ○

```
if a3 >= 0.5:
  yhat = 1
else:
  yhat = 0
```

Andrew Ng

Feature vectors

temperature (Celsius)	duration (minutes)	Good coffee? (1/0)
200.0	17.0	1
425.0	18.5	0
…	…	…

```
x = np.array([[200.0, 17.0]])
```

```
[[200.0, 17.0]]
```

Why?

Andrew Ng

Note about numpy arrays

3 columns

2 rows $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

2 x 3 matrix

```
x = np.array([[1, 2, 3],
              [4, 5, 6]])
```

```
[[1, 2, 3],
 [4, 5, 6]]
```

2D array

2 x 3

4 rows $\begin{bmatrix} 0.1 & 0.2 \\ -3 & -4 \\ -.5 & -.6 \\ 7 & 8 \end{bmatrix}$

2 columns

4 x 2 matrix

```
x = np.array([[0.1, 0.2],
              [-3.0, -4.0,],
              [-0.5, -0.6,],
              [7.0, 8.0,]])
```

```
[[0.1, 0.2],
 [-3.0, -4.0,],
 [-0.5, -0.6,],
 [7.0, 8.0,]]
```

4 x 2

1 x 2

2 x 1

Andrew Ng

Note about numpy arrays

```
x = np.array([[200, 17]])
```

$$\begin{bmatrix} 200 & 17 \end{bmatrix}$$

1 x 2

```
x = np.array([[200],
              [17]])
```

$$\begin{bmatrix} 200 \\ 17 \end{bmatrix}$$

2 x 1

```
x = np.array([200,17])
```

1D
"Vector"

Andrew Ng

Feature vectors

temperature (Celsius)	duration (minutes)	Good coffee? (1/0)
200.0	17.0	1
425.0	18.5	0
…	…	…

```
x = np.array([[200.0, 17.0]])
```

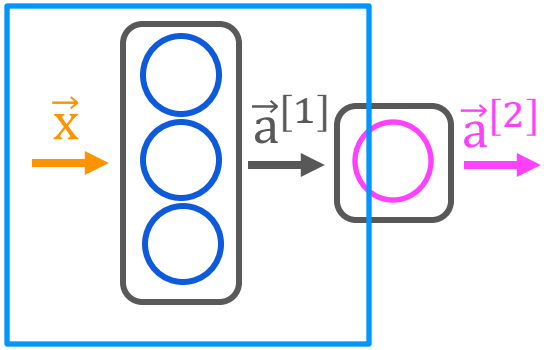
```
[[200.0, 17.0]]
```

1 x 2

$$\begin{bmatrix} 200.0 & 17.0 \end{bmatrix}$$

Andrew Ng

Activation vector




```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

[[0.2, 0.7, 0.3]] 1 x 3 matrix

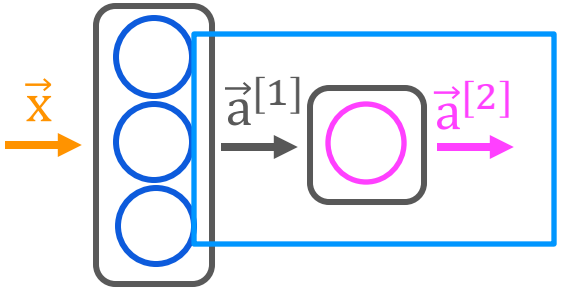
```
tf.Tensor([[0.2 0.7 0.3]], shape=(1, 3), dtype=float32)
```

```
a1.numpy()
```

```
array([[1.4661001, 1.125196 , 3.2159438]], dtype=float32)
```

Andrew Ng

Activation vector




```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

[[0.8]]

1 x 1

```
tf.Tensor([[0.8]], shape=(1, 1), dtype=float32)
```

```
a2.numpy()
```

```
array([[0.8]], dtype=float32)
```

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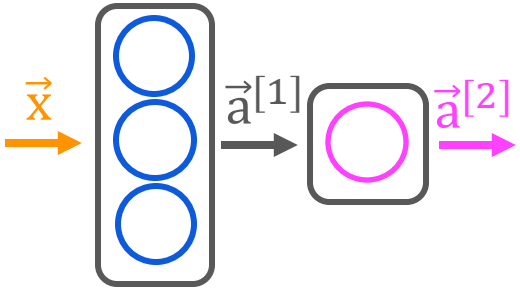
What you saw earlier




```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation="sigmoid")
a1 = layer_1(x)

layer_2 = Dense(units=1, activation="sigmoid")
a2 = layer_2(a1)
```

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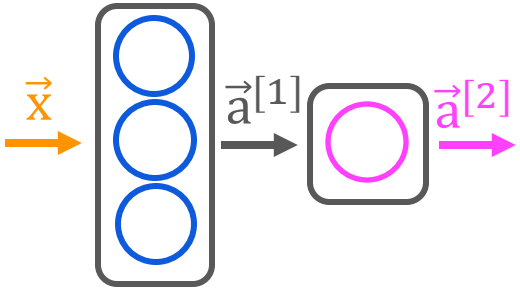
Building a neural network architecture



		y
200	17	1
120	5	0
425	20	0
212	18	1

layer1

```
layer_1 = Dense(units=3, activation="sigmoid")
layer_2 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2])
```

layer2

```
x = np.array([[200.0, 17.0],
              [120.0, 5.0],
              [425.0, 20.0],
              [212.0, 18.0])
```

4 x 2

targets

```
y = np.array([1,0,0,1])

model.compile(...)
model.fit(x,y)

model.predict(x_new)
```

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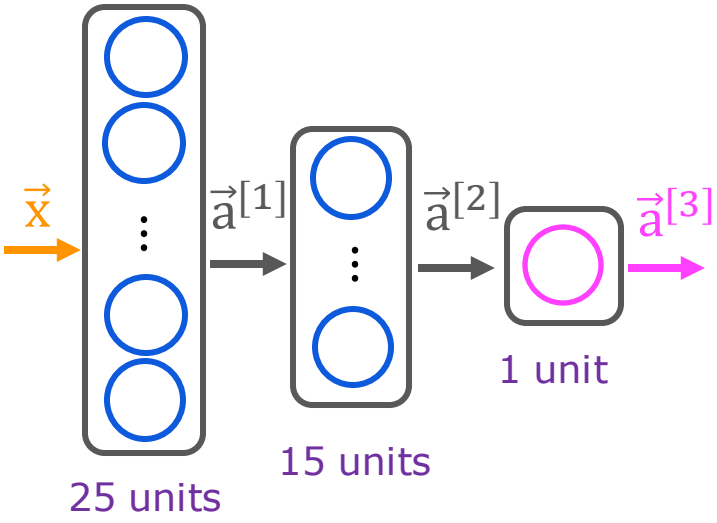
Digit classification model




```
layer_1 = Dense(units=25, activation="sigmoid")
layer_2 = Dense(units=15, activation="sigmoid")
layer_3 = Dense(units=1, activation="sigmoid")

model = Sequential([layer_1, layer_2, layer_3]

model.compile(...)

x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184])
y = np.array([1,0])

model.fit(x,y)

model.predict(x_new)
```

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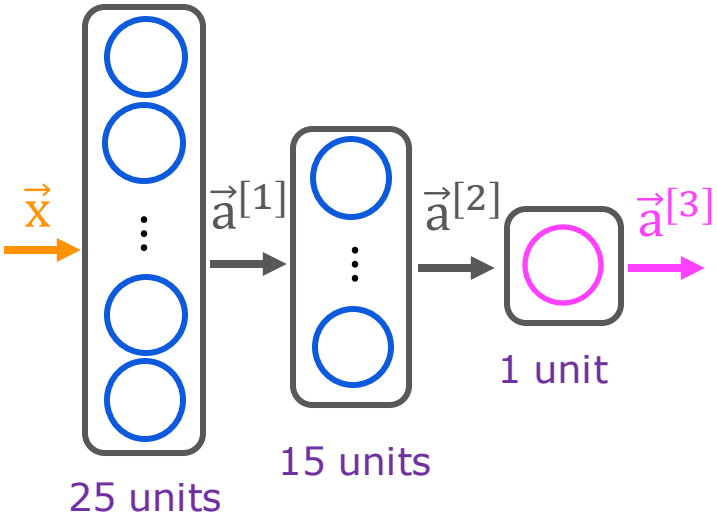
Digit classification model




```
model = Sequential([
  Dense(units=25, activation="sigmoid"),
  Dense(units=15, activation="sigmoid"),
  Dense(units=1, activation="sigmoid")])

model.compile(...)

x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184])
y = np.array([1,0])

model.fit(x,y)

model.predict(x_new)
```

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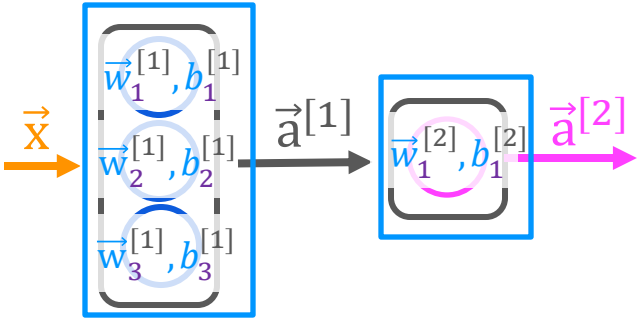
Neural network implementation in Python

Forward prop in a single layer

forward prop (coffee roasting model)



$$a_1^{[2]} = g(\vec{w}_1^{[2]} \cdot \vec{a}^{[1]} + b_1^{[2]})$$

```
w2_1 = np.array([-7, 8])
b2_1 = np.array([3])
z2_1 = np.dot(w2_1,a1)+b2_1
a2_1 = sigmoid(z1_1)
```

```
x = np.array([200, 17])
```

1D arrays

$$a_1^{[1]} = g(\vec{w}_1^{[1]} \cdot \vec{x} + b_1^{[1]})$$

```
w1_1 = np.array([1, 2])
b1_1 = np.array([-1])
z1_1 = np.dot(w1_1,x)+ b
a1_1 = sigmoid(z1_1)
```

$$a_2^{[1]} = g(\vec{w}_2^{[1]} \cdot \vec{x} + b_2^{[1]})$$

```
w1_2 = np.array([-3, 4])
b1_2 = np.array([1])
z1_2 = np.dot(w1_2,x)+b
a1_2 = sigmoid(z1_2)
```

$$a_3^{[1]} = g(\vec{w}_3^{[1]} \cdot \vec{x} + b_3^{[1]})$$

```
w1_3 = np.array([5, -6])
b1_3 = np.array([2])
z1_3 = np.dot(w1_3,x)+b
a1_3 = sigmoid(z1_3)
```

```
a1   = np.array([a1_1, a1_2, a1_3])
```

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For loops vs. vectorization

```
x = np.array([200, 17])

W = np.array([[1, -3, 5],
              [-2, 4, -6]])
b = np.array([-1, 1, 2])

def dense(a_in,W,b):
  units = W.shape[1]
  a_out = np.zeros(units)
  for j in range(units):
    w = W[:,j]
    z = np.dot(w,x) + b[j]
    a_out[j] = g(z)
 return a_out
```

[1,0,1]

```
X = np.array([[200, 17]])
```

2D array

```
W = np.array([[1, -3, 5],
              [-2, 4, -6]])
```

same

```
B = np.array([[-1, 1, 2]])
```

1x3 2D array

all 2D arrays

```
def dense(A_in,W,B):
  Z = np.matmul(A_in,W) + B
  A_out = g(Z)
  return A_out
```

vectorized

matrix multiplication

[[1,0,1]]

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Dot products

example

$$\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix}$$

$$z = (1 \times 3) + (2 \times 4)$$

$$3 + 8$$

$$11$$

in general

$$\begin{bmatrix}\uparrow\\\vec{a}\\\downarrow\end{bmatrix} \cdot \begin{bmatrix}\uparrow\\\vec{w}\\\downarrow\end{bmatrix}$$

$$z = \vec{a} \cdot \vec{w}$$

transpose

$$\vec{a} = \begin{bmatrix}1\\2\end{bmatrix}$$

$$\vec{a}^T = \begin{bmatrix}1 & 2\end{bmatrix}$$

vector vector multiplication

$$\underset{1\times 2}{\begin{bmatrix}\leftarrow & \vec{a}^T & \rightarrow\end{bmatrix}} \underset{2\times 1}{\begin{bmatrix}\uparrow\\\vec{w}\\\downarrow\end{bmatrix}}$$

$$z = \vec{a}^T \vec{w}$$

equivalent

useful for understanding matrix multiplication

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Vector matrix multiplication

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{a}^T = [1 \quad 2] \qquad \mathbf{W} = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix} \qquad \mathbf{Z} = \vec{a}^T \mathbf{W} \quad [\leftarrow \quad \vec{a}^T \quad \rightarrow] \begin{bmatrix} \uparrow & \uparrow \\ \vec{w}_1 & \vec{w}_2 \\ \downarrow & \downarrow \end{bmatrix}$$

$$\mathbf{Z} = \begin{bmatrix} \vec{a}^T \vec{w}_1 & a^T \vec{w}_2 \end{bmatrix}$$

$$(1 * 3) + (2 * 4) \qquad\qquad (1 * 5) + (2 * 6)$$

$$3 + 8 \qquad\qquad 5 + 12$$

$$11 \qquad\qquad 17$$

$$\mathbf{Z} = [11 \quad 17]$$

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matrix matrix multiplication

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$$

$$\mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix} \quad \mathbf{Z} = \mathbf{A}^T\mathbf{W} = \begin{bmatrix} \leftarrow & \vec{a}_1^T & \rightarrow \\ \leftarrow & \vec{a}_2^T & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow \\ \vec{w}_1 & \vec{w}_2 \\ \downarrow & \downarrow \end{bmatrix}$$

rows columns

row 1 col 1 row 1 col 2

$$= \begin{bmatrix} \vec{a}_1^T \vec{w}_1 & \vec{a}_1^T \vec{w}_2 \\ \vec{a}_2^T \vec{w}_1 & \vec{a}_2^T \vec{w}_2 \end{bmatrix}$$

row 2 col 1 row 2 col 2

$(-1 \times 3) + (-2 \times 4)$

$-3 + -8$

-11

$(-1 \times 5) + (-2 \times 6)$

$-5 + -12$

-17

$$= \begin{bmatrix} 11 & 17 \\ -11 & -17 \end{bmatrix}$$

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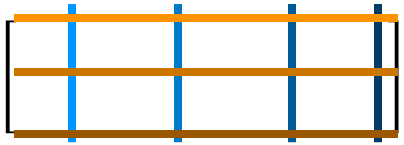
Matrix multiplication rules

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad \mathbf{A}^T = \begin{array}{c} \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \end{array} \begin{array}{l} \vec{a}_1^{\,T} \\ \vec{a}_2^{\,T} \\ \vec{a}_3^{\,T} \end{array} \mathbf{W} = \begin{array}{c} \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \\ \begin{array}{cccc} \vec{w}_1 & \vec{w}_2 & \vec{w}_3 & \vec{w}_4 \end{array} \end{array}$$

$$Z = \mathbf{A}^T \mathbf{W} =$$



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Matrix multiplication rules

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad \mathbf{Z} = \mathbf{A}^T\mathbf{W} = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

3 by 4 matrix

$$\vec{a}_1^T \vec{w}_1 = (1 \times 3) + (2 \times 4) = 11$$

row 3 column 2

$$\vec{a}_3^T \vec{w}_2 = (0.1 \times 5) + (0.2 \times 6) = 1.7$$

$$0.5 + 1.2$$

row 2 column 3?

$$\vec{a}_2^T \vec{w}_3 = (-1 \times 7) + (-2 + 8) = -23$$

$$-7 + -16$$

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Matrix multiplication in NumPy

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad \mathbf{Z} = \mathbf{A}^T\mathbf{W} = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

```
A=np.array([1,-1,0.1],
           [2,-2,0.2]])
```

```
W=np.array([3,5,7,9],
           [4,6,8,0])
```

```
Z = np.matmul(AT,W)
```

or

```
Z = AT @ W
```

```
AT=np.array([1,2],
            [-1,-2],
            [0.1,0.2])
```

```
AT=A.T
```

transpose

result

```
[[11,17,23,9],
 [-11,-17,-23,-9],
 [1.1,1.7,2.3,0.9]
]
```

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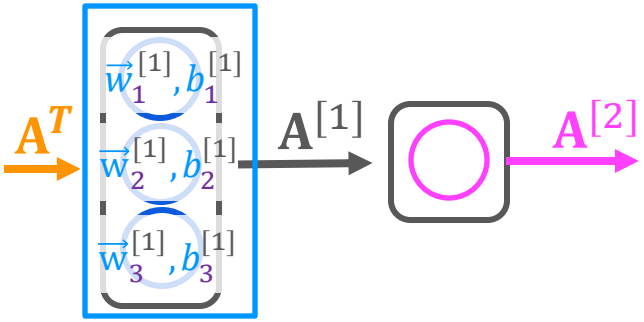
Dense layer vectorized



$$\mathbf{A}^T = \begin{bmatrix} 200 & 17 \end{bmatrix}$$

1 x 2

$$\mathbf{W} = \begin{bmatrix} \boxed{\begin{matrix} 1 \\ -2 \end{matrix}} & \boxed{\begin{matrix} -3 \\ 4 \end{matrix}} & \boxed{\begin{matrix} 5 \\ -6 \end{matrix}} \end{bmatrix}$$

2 x 3

$$\vec{\mathbf{b}} = \begin{bmatrix} -1 & 1 & 2 \end{bmatrix}$$

1 x 3

$$\mathbf{Z} = \mathbf{A}^T\mathbf{W} + \mathbf{B}$$

$$\begin{array}{c} \begin{bmatrix} \boxed{165} & \boxed{-531} & \boxed{900} \end{bmatrix} \\ z_1^{[1]} \quad z_2^{[1]} \quad z_3^{[1]} \end{array}$$

$$\mathbf{A} = g(\mathbf{Z})$$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

```
AT = np.array([[200, 17]])
W = np.array([[1, -3, 5],
              [-2, 4, -6])
b = np.array([[-1, 1, 2]])

def dense(AT,W,b,g):
  z = np.matmul(AT,W) + b
  a_out = g(z)
  return a_out
```

[[1,0,1]]

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